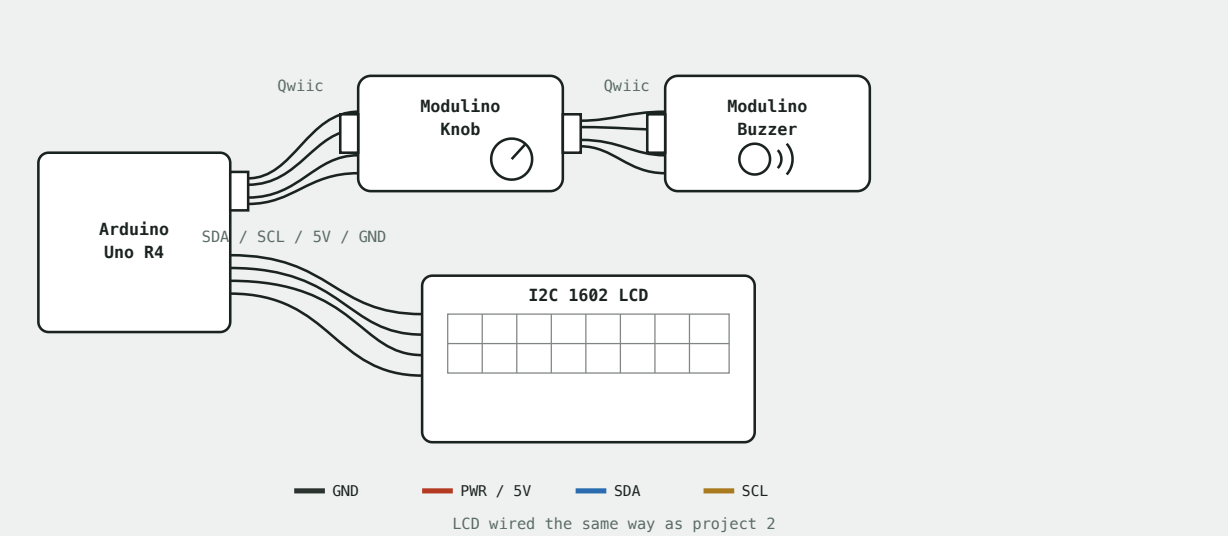
KIT-PROJECTS · 03 · WIRING & CODE REFERENCE

Knob-controlled buzzer piano

Turn the Modulino Knob to pick one of 8 notes across an octave; the LCD from project 2 shows the note's name. Click the knob's built-in button to play it on the Modulino Buzzer.

Board · Arduino Uno R4 **Modules** · Modulino Knob, Modulino Buzzer, I2C 1602 LCD
New concepts · rotary input, `tone()`, lookup arrays



Two Modulino modules chain on one Qwiic cable in either order; the LCD is wired exactly as in **project 2** (header pins, 5V) and isn't re-explained here.

PIN & CONNECTION REFERENCE

Component	Connects to
Modulino Knob	Qwiic/I2C bus
Modulino Buzzer	Qwiic/I2C bus
I2C 1602 LCD	VCC→5V, GND→GND, SDA→SDA, SCL→SCL (same as project 2)

REQUIRED LIBRARIES

- `Arduino_Modulino` — pulls in the sensor-vendor dependencies automatically
- `LiquidCrystal_I2C` — Library Manager

CODE REFERENCE

Input — knob direction → note index

`getDirection()` is +1 / −1 per click; adding 8 before `% 8` wraps both ways.

```
// --- Variables (top of sketch) ---
const int freqs[8] = {262,294,330,349,
                      392,440,494,523};
int index = 0;

// --- Inside loop() ---
index = (index + knob.getDirection() + 8)
% 8;
```

Output — play a note

`tone()` takes a frequency in Hz and a duration in ms.

```
// --- Inside loop() ---
if (knob.isPressed()) {
  buzzer.tone(freqs[index], 150);
}
```

TRY IT YOURSELF

- Add a second octave by doubling the arrays and changing every `8` in the index line to `16`.
- Sustain the note while the button is held, instead of one short beep.